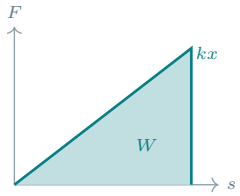
CALCULUS ON ONE PAGE

Slopes as instants, areas as totals

Triangles give work and impulse without calculus; limits give instantaneous rates and measurable totals.

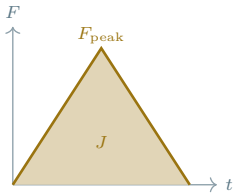
1 / WITHOUT CALCULUS

Areas under straight-line graphs are measurable totals



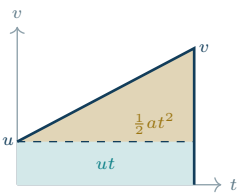
spring

$W = \frac{1}{2}x \cdot kx$



bat on ball

$J = \frac{1}{2}F_{\text{peak}}\Delta t$



constant acceleration

$s = ut + \frac{1}{2}at^2$

A force that grows steadily with distance, $F = kx$, draws a triangle on a graph of force against distance, and its area is the work, which is the energy stored in the spring. A force that rises and falls steadily in a collision draws a triangle against time, and its area is the impulse. A constant acceleration draws a sloping velocity line, and the area under it is a rectangle ut and a triangle $\frac{1}{2}at^2$.

$$W = \tfrac{1}{2}kx^2$$
$$J = \tfrac{1}{2}F_{\text{peak}}\Delta t$$
$$s = ut + \tfrac{1}{2}at^2$$

The units say what each area is, newtons times metres, newtons times seconds, or metres per second times seconds. Compressing a spring of 200 N/m by 0.10 m needs a force that grows to 20 N, so $W = \frac{1}{2}\cdot 20\cdot 0.10 = 1.0$ J, and from 0.05 m to 0.10 m the trapezoid $\frac{1}{2}(10+20)(0.05) = 0.75$ J. A bat that reverses a 0.15 kg ball from -40 m/s to 40 m/s gives $J = 0.15 \cdot 80 = 12$ N s, which a triangular pulse of 1 ms achieves with a peak force of 24 kN. With $u = 5$ m/s, $a = 2$ m/s^2 and $t = 4$ s the areas are 20 m and 16 m, so $s = 36$ m.

2 / DIFFERENTIATION

The slope is the instantaneous measure

Curved graphs have no triangle to measure, so the slope at one instant is the limit of slopes over shrinking intervals. For $s = t^3$ at $t = 2$:

h	$\frac{s(2+h)-s(2)}{h}$
1	19
0.1	12.61
0.01	12.0601
0.001	12.006001
$\to 0$	$12 = 3t^2$ at $t = 2$

The limit is the instantaneous speed $v = \frac{ds}{dt} = 3t^2 = 12$ m/s, and the same step gives $a = \frac{dv}{dt}$, $F = \frac{dp}{dt}$, a current $\frac{dQ}{dt}$ and an emf $-\frac{d\Phi}{dt}$.

3 / INTEGRATION

The sum of strips is the measurable total

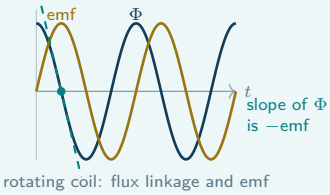
The area under a curve is bracketed by sums of thin rectangles, and the limit is the integral. For $v = 3t^2$ from 0 to 2 s:

n strips	left sum	right sum
2	3.0000	15.0000
4	5.2500	11.2500
8	6.5625	9.5625
16	7.2656	8.7656
100	7.8804	8.1204
∞	8	$8 = t^3\big\vert_0^2$

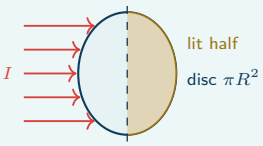
The total is the displacement $s = \int v\,dt = t^3 = 8$ m, and in same way $W = \int F\,dx$, $J = \int F\,dt$ and $\Phi = \int \mathbf{B}\cdot d\mathbf{A}$. For $v = u + at$ integral is $s = ut + \frac{1}{2}at^2$, the rectangle and triangle again.

4 / TWO INSTRUMENTS

A flux is an integral over an area and its rate of change is an emf



rotating coil: flux linkage and emf



sunlight on the Earth

A coil of N turns and area A rotating at ω in a field B has the flux linkage $\Phi = NBA\cos\omega t$, an integral of B over the area, and the emf $-\frac{d\Phi}{dt} = NBA\omega\sin\omega t$ is the slope of that curve. For $N = 100$, $A = 0.01$ m^2, $B = 0.05$ T and 50 Hz the peak is $0.05 \cdot 314.16 = 15.7$ V. Sunlight of irradiance $I = 1361$ W/m^2 meets the lit hemisphere at the angle θ to its surface normal, so the power is $\int I\cos\theta\,dA = \int_0^{\pi/2} I\cos\theta\,2\pi R^2\sin\theta\,d\theta = I\pi R^2$, which is 1.74×10^{17} W for $R = 6.371\times10^6$ m, and spread over the whole sphere the average is $\frac{I}{4} = 340$ W/m^2.

5 / THE TWO OPERATIONS SIDE BY SIDE

Each derivative is read at an instant or a point, and each integral over an interval or a region

field	instantaneous measure (a derivative)	measurable total (an integral)
motion	speed $v = \frac{ds}{dt}$, acceleration $a = \frac{dv}{dt}$	distance $s = \int v\,dt$, change of velocity $\int a\,dt$
mechanics	force $F = \frac{dp}{dt}$	work $W = \int F\,dx$, impulse $J = \int F\,dt$
electromagnetism	emf $-\frac{d\Phi}{dt}$, current $i = \frac{dQ}{dt}$	flux $\Phi = \int \mathbf{B}\cdot d\mathbf{A}$, charge $Q = \int i\,dt$
radiation	irradiance $I = \frac{dP}{dA}$	power $P = \int I\,dA$, energy $\int P\,dt$

ASIDE / TRY THESE

Use $h = 0.0001$ in the table for $s = t^3$ and see 12.0006, and use $n = 1000$ strips to see the sums 7.988 and 8.012 close in on 8.

LINKS TO OTHER SHEETS

The work and impulse sheet and the constitutive laws sheet use these integrals, and the power series sheet builds on the integral of a geometric series.

Convention. s distance, t time, v velocity, Φ flux linkage; the limit tables, the emf and the sunlight integral were checked by computation.

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